

Parallel computing

- * Uses multiple processors to solve problems faster.
- * The objective is to reduce execution time + improve performance.

Applications

- ML
- Weather forecasting, big data.

Performance metrics

- Speedup
- Efficiency.
- Scalability.

Factors Affecting Performance

- Sequential part
- Communication overhead
- Load imbalance
- Synchronization delay.

Amdahl's Law

- * Defines max speed limit, based on fixed problem size.

$$\text{Speedup (S)} = \frac{1}{1 - f + \frac{f}{N}} = \frac{1}{(1-f) + \frac{f}{N}}$$

Characteristics of Amdahl's Law

- Sequential bottleneck limits performance improvement.
- Max. speed limit $\Rightarrow$ Even with 1000 processors, speedup cannot exceed 2.
- Fixed problem size.
- Diminishing returns.

$$S(N) = \frac{1}{(1-P) + \frac{P}{N}}$$

$S(N)$ = Speedup using n processors.

P = Fraction of program that can run in parallel.

$1-P$ = Serial portion.

N = Number of processors.

Ex:- $P = 0.9$

$N = 10$

$$S = \frac{1}{(1-0.9) + \frac{0.9}{10}} = \frac{1}{0.1 + 0.09} = 5.26$$

Amdahl's Law Applications

- Multi core processor design.
- Super computers.
- GPU computing.
- Scientific simulations.

Gustafson's Law

* Speedup of a parallel program can increase linearly with the number of processors if the problem size scales with the system.

* For Scalable workload:

* This says instead of using more processors to solve the same problem faster, we can increase the size of the problem, so that system performs more useful work in the same time.

$$S(N) = N - (1 - P)(N - 1)$$

N = Number of processors

Characteristics

- * Scalable problem size.
- * Parallel dominance.
- * Linear speedup possibility.

Applications

- Climate & weather modeling.
- Big data Analytics & AI/ML
- 3D rendering & Animation.
- Space Exploration Simulations.

AWS parallel cluster

* Tool to deploy & manage HPC clusters in AWS.

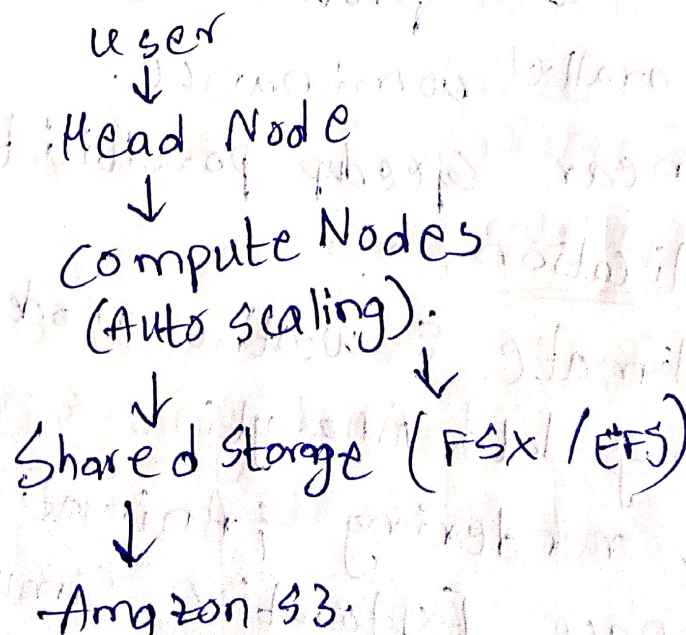
Features

- Automatic creation of compute clusters.
- High speed networking.
- Shared storage.
- Auto scaling.

Architecture components

- Head node (Central controller of the cluster)
- Compute node (performs actual parallel computation).
- Storage system (shared high speed storage)
- Networking (fast communication b/w nodes).

Work flow



Advantages

- Automatic cluster deployment.
- Elastic scaling of compute nodes.
- High performance networking.
- Shared parallel file systems.
- Flexible resource scaling.

Applications

- AI model training.
- Scientific simulations.
- Financial risk analysis.
- Genome sequencing.

Azure Batch

- * Runs many parallel jobs without worrying about servers.

Key features

- Job scheduling & task management.
- Automatic VM provisioning.

Applications

- Video rendering.
- Data Analysis
- Financial risk modeling.

Key components

- Batch Account
- Compute pools
- Job manager
- Task scheduler.

Google Cloud HPC

→ Platform for running HPC workloads using scalable infrastructure.

Key features

- High performance VM's.
- Custom machine types.
- Fast networking.

Key components

- Computer resources.
- Storage resources.
- Networking resources.
- Scheduling systems.

Auto Scaling

* Automatically adjusts resources based on workload.

Types

→ Vertical Scaling

* Increasing CPU or memory of a single machine.

→ Horizontal Scaling

* Adding more nodes to the cluster.

LINPACK BENCHMARK

* This is used to measure the floating point computing performance of HPC systems.

* Solves $Ax=b$ using dense matrices.

* Based on Gaussian elimination algorithm, produced results in FLOP's.

Applications

- AI & ML
- Computer Graphics & Gaming.
- Weather forecasting.
- Medical Imaging.

Working

- Generate a large dense matrix
- Applies Gaussian elimination.
- Apply LU decomposition.
- Solves the system of equations.

Performance Metrics

- R_{max} ⇒ (Actual achieved performance)
- R_{peak} ⇒ Max theoretical performance.

HPC G Benchmark

* High performance Conjugate gradient benchmark.

* measures real-world HPC performance

Characteristics

- Uses sparse matrices
- Implements conjugate gradient algorithm.
- Represents realistic scientific workloads.

Working Principle

- 1.) Construct 3D sparse matrix problem
↓
- 2.) Apply Conjugate Gradient method.
↓
- 3.) Sparse matrix - vector multiplication
↓
- 4.) Gauss-Seidel Smoothing.
↓
- 5.) Measure performance & Report FLOPS.

<u>LINPACK</u>	<u>HPCG</u>
* Dense computation	* Sparse computation.
* CPU-focused	* Memory + Network focused.
* Peak performance	* Real-world performance

Cloud HPC pricing models

- Pay-as-you-go.
 - Reserved instance.
 - Spot instance
- } models.

Pay-as-you-go

- * Users pay only for the resources they consume.

Reserved instance model

- * Users reserve computing resources for a longer period at discounted rates.

Spot instance

- * Unused cloud resources are offered at very low prices but can be terminated any time.

Applications of cloud HPC

- Weather forecasting.
- Genomics research
- Financial modeling.

AI Accelerators

- * Designed to efficiently execute neural network computations.

Main Components

→ Processing Units

- * Performs parallel computations

* Units execute,

→ matrix multiplication.

→ Vector operations.

→ Convolution operations.

→ Memory hierarchy

→ Interconnection network.

→ Tensor processing units.

→ Control unit.

Applications

→ Computer vision.

→ Natural Language Processing.

→ Autonomous vehicles.

→ Robotics.

GPU Acceleration

- * GPU acceleration improves the training performance of machine learning models by enabling massively parallel computation of complex mathematical operations.

Applications

- Image recognition
- Natural language Processing
- Speech recognition Systems

Cryptographic Accelerators

- * These are essential for modern security systems, enabling fast & efficient encryption

Ex:- CPU-based acceleration

- FPGA
- ASIC

Importance

- Speed up transactions
- Improve scalability
- Reduce CPU Load
- Enhance security

Types of Cryptographic Accelerators

Dedicated Hardware

* High security & performance.

Ex:- HSM

FPGA

ASIC

Integrated Accelerators

→ Build into:

* CPU

* GPU

* Network Interface Cards.

→ GPU Acceleration

→ FPGA Acceleration

→ ASIC Acceleration

→ CPU Instruction Based Acceleration.

Advantages

→ High speed & performance.

→ Energy Efficient.

→ Improved security.

→ Scalable for large systems.

Applications

- Secure Web Communication (SSL/TLS).
- Banking & Financial Systems.
- Block Chain & Crypto currencies.

* Cryptographic accelerators allows cloud systems to handle large-scale secure transactions efficiently, so this plays a critical role in cloud security.

Cryptographic Functions

- AES (Advanced Encryption Standard).
- RSA (Rivest-Shamir-Adleman)
- ECC (Elliptic Curve Cryptography)
- HASH (SHA-2, SHA-3)
- RNG (Random Number Generator).

Working Principle

- User sends request.
- Request goes to Crypto Accelerator.
- Hardware engines process operations.
- Processed data returned securely.
- Sent to cloud service.

Advantages

- Faster encryption & decryption.
- Reduced CPU workload.
- Enhanced security.

Hardware Accelerators in video processing

* Used to process high-speed video data in real time.

1) Video input stage

* Process begins with capturing raw video data.

Sources,

→ Security Cameras, Drones, Smartphones, Self driving cars.

2) Video processing tasks

→ Object Detections

→ Pose

→ Object Tracking

→ Heatmap / Scene Analysis.

3) Note: - Accelerators in real-time video processing are GPU, FPGA, VPU, ASIC.

4) Results are returned to CPU

5) Final output is generated.

Applications

- Object Detection (CPU)
- Face Recognition (CPU)
- Tracking & segmentation (CPU + FPGA).
- Super-Resolution (CPU).

Automotive systems

- Perception systems
- Autonomous Driving.
- Driver monitoring.

Applications

- Real time video streaming.
- Surveillance systems
- Video Analytics.